Bluetooth devices from the Internet for general IoT use cases, then you need a gateway, which has some kind of adaptation scheme (an API) so that requests can be expressed using a TCP/IP based protocol and then translated by the gateway into the appropriate Bluetooth requests and sent to a connected Bluetooth device.

To use Bluetooth in an IoT context, you usually need something which can convert between some application protocols that run over TCP/IP and the Bluetooth protocols. That is generally called a Bluetooth Internet gateway (Figure 2) (BIG), which is classified as middleware.

Bluetooth Internet gateway requirements

These days we have the older Bluetooth BR/EDR and the newer Bluetooth LE to choose from (Figure 3). Bluetooth LE can be used in various ways including connection-oriented communication and connectionless and also for creating large mesh networks of Bluetooth devices. Therefore, one has to first consider what type of Bluetooth technology needs to be supported.

For this project, Bluetooth LE was used since support for Bluetooth mesh was not the priority and there was no intention to support the older Bluetooth BR/EDR either.

Bluetooth LE devices can work in various ways. Connectionless communication involves a procedure known as advertising where small packets of data are broadcast and received by a device in range (scanning). This mechanism is used for facilitating device discovery. Typically, scanning devices like smartphones find and choose a device with a little help from the user and then establish a connection. After this, communication takes place over the connection. And this is required to be done by the gateway on behalf of a TCP/IP client of some sort.

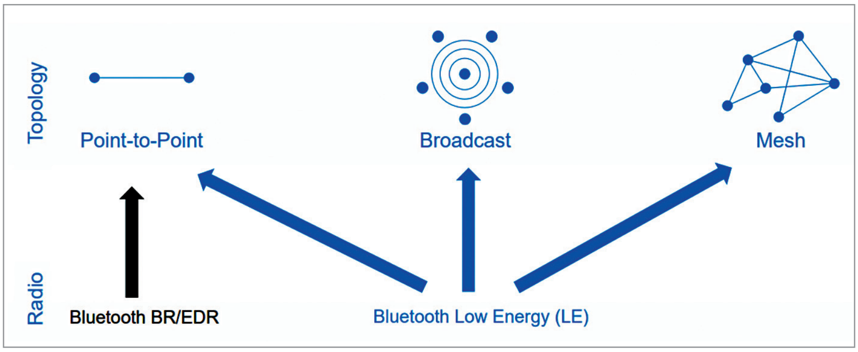

Figure 3: The older Bluetooth BR/EDR and the newer Bluetooth LE

Short Name	Details
Bluetooth device types	The gateway act as **GAP Central** and support connectable Bluetooth devices in the **GAP peripheral** and **GATT server** roles only.
Protocol and Remote Operations	The gateway shall allow applications to invoke Bluetooth operations using a simple and commonly supported TCP/IP-based protocol.
Device Discovery	The gateway must be able to discover advertising Bluetooth devices and report details to clients.
Connect	The gateway must connect to a selected Bluetooth device on request
Service Discovery	The gateway must be able to discover GATT services, characteristics and descriptors implemented by a connected GATT server device
Read	The gateway must read a characteristic value from a connected Bluetooth device on request.
Write	The gateway must write a new value to a characteristic in a connected Bluetooth device on request.
Notify	The gateway must allow characteristic value notifications to be sent from a connected Bluetooth device to a gateway client.

Figure 4: Bluetooth Internet gateway requirements

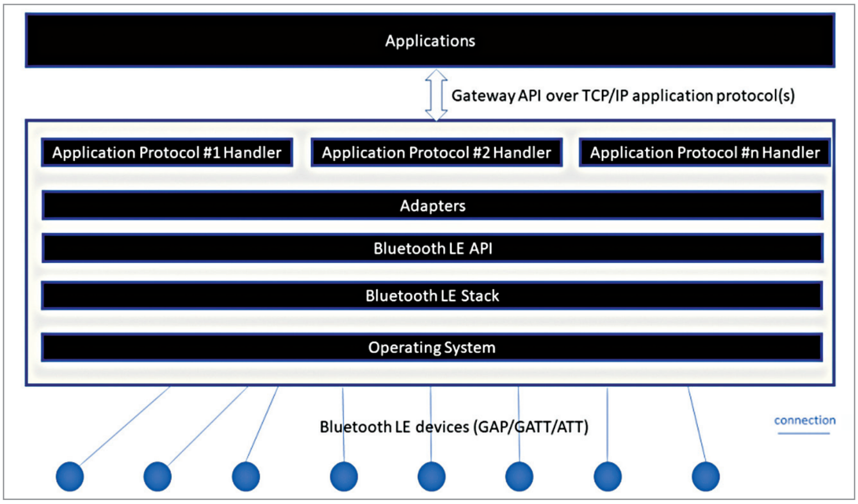

Figure 5: Gateway logical architecture

Connectable devices that advertise are called peripherals. And those that scan and request connections are called central devices.

So, it was decided to support connectable LE devices that act as peripherals with a Bluetooth Internet gateway acting as the central device that would do scanning and connection.

Bluetooth Internet gateway architecture

Adapter components handle requests, which are encoded in TCP/IP